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.1 after a steady approach with zero yaw rate, the rudder is put over to 10° to starboard or port (first execute);

.2 when the heading has changed to 10° off the original heading, the rudder is reversed to 10° to port or starboard (second execute); and

.3 after the rudder has been turned to port/starboard, the ship will continue turning in the original direction with decreasing turning rate. In response

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reversed to 10° to port or starboard (second execute); and

.3 after the rudder has been turned to port/starboard, the ship will continue turning in the original direction with decreasing turning rate. In response to the rudder, the ship should then turn to port/starboard. When the ship has reached a heading of 10° to port/sta rboard of the original course the rudder is again reversed to 10° to starboard/port (third execute).

.7 The first overshoot angle is the additional heading deviation experienced in the zig-zag test following the second execute.

.8 The second overshoot angle is the additional heading deviation experienced in the zig-zag test following the third execute.

.9 The 20°/20° zig-zag test is performed using the procedure given in paragraph 4.2.6 above using 20° rudder angles and 20° change of heading, instead of 10° rudder angles and 10° change of heading, respectively.

.10 Full astern stopping test determines the track reach of a ship from the time an order for full astern is given until the ship stops in the water.

.11 Track reach is the distance along the path described by the midship point of a ship

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SOURCE: MSC.137(76).pdf
.10 Full astern stopping test determines the track reach of a ship from the time an order for full astern is given until the ship stops in the water.

.11 Track reach is the distance along the path described by the midship point of a ship measured from the position at which an order for full astern is given to the position at which the ship stops in the water.

5 S TANDARDS

5.1 The standard manoeuvres should be performed without the use of any manoeuvring aids which are not continuously and readily available in normal operation.

5.2 Conditions at which the standards apply

In order to evaluate the performance of a sh ip, manoeuvring trials should be conducted to both

port and starboard and at conditions specified below:

.1 deep, unrestricted water;

.2 calm environment;

.3 full load (summer load line draught), even keel condition; and

.4 steady approach at the test speed.

5.3 Criteria *

The manoeuvrability of the ship is considered satisfactory if the following criteria are complied with:

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SOURCE: MSC.137(76).pdf
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5.3 Criteria *

The manoeuvrability of the ship is considered satisfactory if the following criteria are complied with:

.1 Turning ability

The advance should not exceed 4.5 ship le ngths (L) and the tactical diameter should not exceed 5 ship lengths in the turning circle manoeuvre.

.2 Initial turning ability

With the application of 10° rudder angl e to port/starboard, the ship should not have travelled more than 2.5 ship lengths by the time the heading has changed by 10° from the original heading.

.3 Yaw-checking and course-keeping abilities

.1 The value of the first overshoot angle in the 10°/10° zig-zag test should not exceed:

.1 10° if L/V is less than 10 s;

.2 20° if L/V is 30 s or more; and

.3 (5 + 1/2(L/V)) degrees if L/V is 10 s or more, but less than 30 s,

where L and V are expressed in m and m/s, respectively.

.2 The value of the second overshoot angl e in the 10°/10° zig-zag test should not exceed:

.1 25°, if L/V is less than 10 s;

.2 40°, if L/V is 30 s or more; and

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SOURCE: MSC.137(76).pdf
where L and V are expressed in m and m/s, respectively.

.2 The value of the second overshoot angl e in the 10°/10° zig-zag test should not exceed:

.1 25°, if L/V is less than 10 s;

.2 40°, if L/V is 30 s or more; and

.3 (17.5 + 0.75(L/V))°, if L/V is 10 s or more, but less than 30 s.

.3 The value of the first overshoot angle in the 20°/20° zig-zag test should not exceed 25°.

.4 Stopping ability

The track reach in the full astern stoppi ng test should not exceed 15 ship lengths. However, this value may be modified by the Administration where ships of large displacement make this criterion impracticable, but should in no case exceed 20 ship lengths.

* For ships with non-conventional steering and propulsion systems, the Administration may permit the use of comparative steering angles to the rudder angles specified by this Standard.

6 A DDITIONAL CONSIDERATIONS

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RESOLUTION MSC.137(76)
(adopted on 4 December 2002)

6 A DDITIONAL CONSIDERATIONS

6.1 In case the standard trials are conducted at a condition different from those specified in paragraph 5.2.3, necessary corrections should be made in accordance with the guidelines contained in the Explanatory notes to the Standards for ship manoeuvrability, developed by the Organization.*

6.2 Where standard manoeuvres indicate dynamic instability, alternative tests may be conducted to define the degree of instability. Guidelin es for alternative tests such as a spiral test or pull-out manoeuvre are included in the Explanatory notes to the Standards for ship manoeuvrability, referred to in paragraph 6.1 above.

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Refer to MSC/Circ.1053 on Explanatory notes to the Standards for ship manoeuvrability.
RESOLUTION MSC.137(76)